## 2.2.2 Resistors in series and parallel

An electric circuit may contain a number of resistors which can be connected in different ways. For each type of circuit, we can calculate the equivalent resistance produced by a group of individual resistors.

### Resistors in series

When two or more resistors are connected end to end, they are said to be in series. The resistors could be simple resistors or bulbs or heating elements or other devices. Figure 2.9 (a) shows three resistors $R_1$, $R_2$ and $R_3$ connected in series.

The amount of charge passing through resistor $R_1$ must also pass through resistors $R_2$ and $R_3$ since the charges cannot accumulate anywhere in the circuit. Due to this reason, the current I passing through all the three resistors is the same. According to Ohm's law, if same current pass through different resistors of different values, then the potential difference across each resistor must be different. Let $V_1$, $V_2$ and $V_3$ be the potential difference (voltage) across each of the resistors $R_1$, $R_2$ and $R_3$ respectively, then we can write $V_1 = IR_1$, $V_2 = IR_2$ and $V_3 = IR_3$. But the total voltage $V$ is equal to the sum of voltages across each resistor.

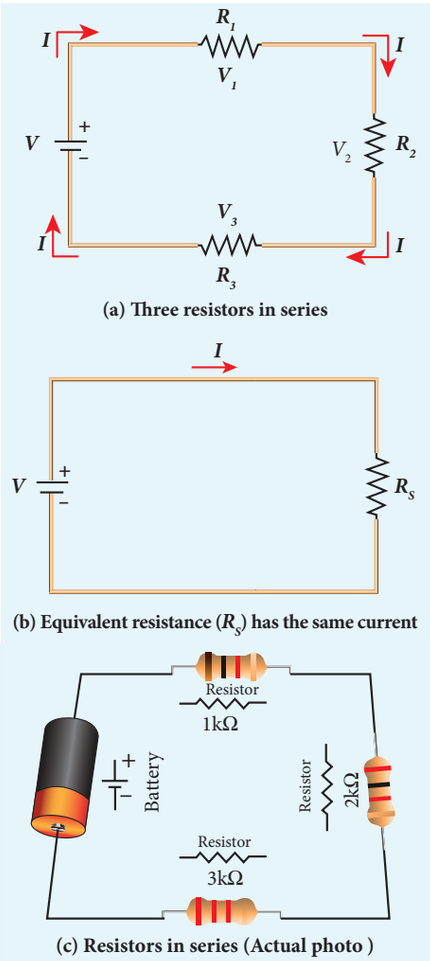



**Figure 2.9** Resistors in series

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 \quad (2.21)$$

$$V = I\,(R_1 + R_2 + R_3)$$

$$V = I.R_S \quad (2.22)$$

where $R_S$ is the equivalent resistance,

$$R_S = R_1 + R_2 + R_3 \quad (2.23)$$

When several resistances are connected in series, the total or equivalent resistance is the sum of the individual resistances as shown in the Figure 2.9 (b).

**Note:** The value of equivalent resistance in series connection will be greater than each individual resistance.

### EXAMPLE 2.8

Calculate the equivalent resistance for the circuit which is connected to 24 V battery and also find the potential difference across 4 Ω and 6 Ω resistors in the circuit.

